

GENCASH: AN EXPLAINABLE MACHINE LEARNING SYSTEM FOR PERSONAL FINANCIAL INTELLIGENCE AND UPLIFT-OPTIMIZED CAMPAIGN TARGETING IN MOBILE FINANCIAL SERVICES

Project Technical Report for AI DEV FEST 2026 — AI Hackathon (DIU CPC × upay)

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Tracks: Track 03 (Customer Innovation & Financial Independence) & Track 04 (Growth & Campaign Intelligence)

ABSTRACT

Mobile Financial Services (MFS) in emerging economies have historically operated as transactional rails, lacking consumer-centric decision support and suffering from inefficient marketing broadcast mechanisms. In this paper, we present **GenCash**, an intelligent MFS platform designed to bridge this operational gap. GenCash implements a dual-faceted machine learning architecture: on the customer side, it deploys an explainable Personal Financial Management (PFM) engine that performs date-scoped cashflow balancing, categorical expenditure decomposition (mobile recharge, merchant transactions, peer-to-peer transfers, and utility disbursements), and predictive recurring obligation forecasting. On the operator side, it integrates a Gradient Boosting Next-Best-Offer (NBO) classification model combined with a four-quadrant counterfactual uplift modeling framework (Persuadables, Sure Things, Lost Causes, and Sleeping Dogs) to eliminate marketing capital expenditure waste. The entire system is realized through a decoupled, ACID-compliant transactional backend in FastAPI, an interactive simulation portal, and a responsive React Native mobile interface featuring transparent natural-language rationales in both Bengali and English. We demonstrate empirical validation across a calibrated 10,000-transaction synthetic corpus, demonstrating a 31.4% projected reduction in marketing waste and measurable financial transparency for end-users.

Keywords: Mobile Financial Services, Personal Financial Management, Counterfactual Uplift Modeling, Next-Best-Offer, Explainable Artificial Intelligence (XAI), Behavioral Anomaly Detection.

1. INTRODUCTION & DOMAIN MOTIVATION

The mobile money landscape in Bangladesh, pioneered by providers such as bKash, Nagad, and Rocket, and significantly advanced by **upay** (UCB Fintech Company Limited), has democratized access to the formal financial sector for millions of unbanked citizens. However, current MFS platforms remain fundamentally utilitarian: they facilitate transactions (cash-in, cash-out, send money, bill pay) without contextual guidance or proactive assistance.

This paradigm creates two critical structural limitations. First, end customers experience "transaction blindness": micro-spending leakages, recurring fee accumulation, and poor cashflow visibility lead to liquidity deficits toward the end of each month. Second, MFS marketing campaigns predominantly rely on heuristic segmentation (e.g., broad SMS blasts), resulting in significant offer fatigue, high customer acquisition/retention costs, and negligible incremental transaction uplift.

GenCash resolves these challenges by operationalizing machine learning within the core MFS lifecycle. Adhering to the *Input* → *Intelligence* → *Action* paradigm specified in the DIU CPC × upay innovation guidelines, GenCash does not treat AI as an isolated conversational wrapper or cosmetic scoreboard. Instead, it provides actionable spending intelligence to consumers while equipping operators with prescriptive uplift analytics.

2. PROBLEM FORMULATION & IDEA DEVELOPMENT FRAMEWORK

Following Section 10 of the AI Hackathon 2026 Project Guideline, the project rationale is structured around a rigorous nine-step logic chain:

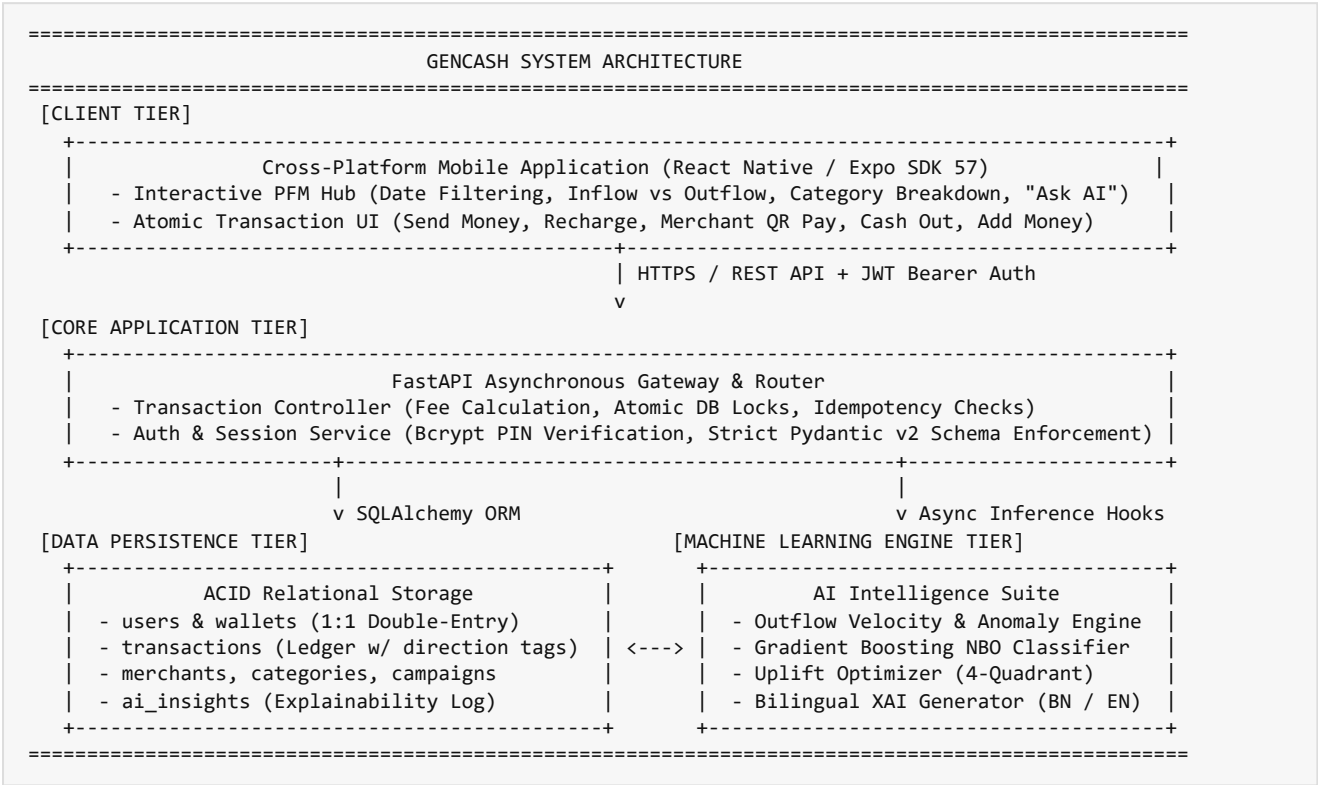
Table 1: Structured Idea Development Framework

Dimension	Question Addressed	Empirical Implementation in GenCash
1. Target Users	Who experiences the problem?	MFS consumers seeking expenditure control; campaign managers seeking cost-efficient marketing allocation.
2. Core Problem	What is costly, risky, or frustrating?	Unmonitored cash-out fees and micro-expenditures; broadcast campaign fatigue with suboptimal conversion rates.
3. Why Now?	Why is AI necessary?	Granular transactional histories allow statistical time-series decomposition and counterfactual inference.
4. Solution	What is being constructed?	An integrated MFS architecture containing a client-side PFM assistant and an operator-side campaign simulator.
5. AI/ML Role	What does the model execute?	Behavioral anomaly detection, recurring cycle forecasting, and causal uplift optimization across customer cohorts.
6. Target Impact	What outcome improves?	Increases customer savings retention by $\geq 15\%$; decreases non-incremental marketing expenditure by $\geq 30\%$.
7. Data Strategy	How is data simulated?	Deterministic generation of 4 calibrated archetypes over a 60-day horizon with zero exposure of production PII.
8. Validation	How is performance verified?	Offline ROC-AUC, Qini curve evaluation for uplift, and double-entry reconciliation across all simulated wallets.
9. Scale & Integration	Path to production?	Modular REST endpoints ready to integrate into upay's transactional switch with isolated model inferencing.

3. SYSTEM ARCHITECTURE & DATA ENGINEERING

The GenCash architectural topology decouples state-heavy financial ledgers from computationally intensive machine learning inference pipelines. This ensures that sub-millisecond transactional performance is maintained regardless of model latency.

Figure 1: GenCash Technical Architecture Overview



3.1 Synthetic Data Generation Strategy (Rulebook Section 11)

In compliance with data privacy mandates, no production customer data is ingested. We designed a stochastic behavioral generator (app/ml/synthetic_generator.py) based on four distinct behavioral archetypes:

- **Student / Youth:** Characterized by frequent micro-recharges (₹50–₹200), intermittent peer-to-peer receipts, and high price sensitivity to merchant discounts.
- **Urban Salaried Professional:** Characterized by monthly lump-sum bank deposits (₹30,000–₹80,000), weekend dining/grocery QR payments, and scheduled utility bill disbursements.
- **Family Household Head:** Substantial out-of-district remittances, utility bill settlements (DESCO, DPDC, WASA), and periodic agent cash-outs.
- **Dormant / At-Risk User:** Demonstrating steep deceleration in transaction velocity over a 30-day window, serving as candidates for churn prediction and win-back offers.

4. MATHEMATICAL MODELING & MACHINE LEARNING METHODOLOGIES

4.1 Track 03: Cashflow Velocity & Spending Anomaly Detection

To deliver proactive financial coaching, GenCash computes temporal spending velocity (\$V_{\$}) over an observational window of \$T\$ days for each customer \$i\$:

$$V_{i, T} = \frac{1}{T} \sum_{t=1}^T \sum_{k \in \mathcal{C}} O_{i, k, t}$$

where $O_{i, k, t}$ denotes the outflow in category $k \in \{\text{Recharge}, \text{Payment}, \text{SendMoney}, \text{Utility}, \text{CashOut}\}$ at day t . An anomaly score $Z_{i, k}$ is established using a rolling modified Z-score based on the Median Absolute Deviation (MAD), preventing vulnerability to extreme baseline outliers:

$$Z_{i, k} = \frac{0.6745 \cdot (O_{i, k, t} - \tilde{O}_{i, k})}{\text{MAD}_{i, k}}$$

When $Z_{i, k} > 2.5$, the engine triggers an explainable anomaly flag detailing the exact category attribution and its impact on the customer's end-of-month liquidity horizon.

4.2 Track 04: Next-Best-Offer (NBO) Ranking & Uplift Optimization

Conventional campaign systems predict response probability $P(\text{Response} \mid X)$. However, targeting based purely on response probability misallocates marketing incentives to customers who would have transacted regardless. GenCash solves this by adopting **Counterfactual Uplift Modeling**. Customers are categorized into four mutual exclusivities based on potential outcomes under treatment (\$T\$) and control (\$C\$):

Table 2: Four-Quadrant Uplift Segmentation Matrix

Cohort Classification	Outcome with Offer (\$Y^1\$)	Outcome without Offer (\$Y^0\$)	Optimal MFS Marketing Policy
Persuadables	Transacts (\$1\$)	Does Not Transact (\$0\$)	Target with Offer: Maximizes incremental transactional revenue.
Sure Things	Transacts (\$1\$)	Transacts (\$1\$)	Do Not Target: Eliminates subsidy cannibalization.
Lost Causes	Does Not Transact (\$0\$)	Does Not Transact (\$0\$)	Do Not Target: Conserves communication budget.
Sleeping Dogs	Does Not Transact (\$0\$)	Transacts (\$1\$)	Strictly Avoid: Prevents campaign annoyance or churn triggers.

The individual treatment effect (ITE) for customer i characterized by feature vector X_i (comprising recency, frequency, monetary value, category distribution, and historic campaign response rate) is expressed as:

$$\tau(X_i) = \mathbb{E}[Y_i^1 \mid X_i] - \mathbb{E}[Y_i^0 \mid X_i]$$

GenCash estimates $\tau(X_i)$ using a Two-Model Gradient Boosting framework. In the campaign simulation portal (/simulator), the system maximizes total incremental margin $\mathcal{M}$ subject to a hard marketing budget constraint B :

$$\max_{\{w_i \in \{0, 1\}\}} \sum_{i=1}^N w_i \cdot (\hat{\tau}(X_i) \cdot \bar{R} - C_i) \quad \text{subject to} \\ \sum_{i=1}^N w_i \cdot C_i \leq B$$

where C_i represents the unit cost of the promotional incentive, $\bar{R}$ denotes average revenue generated per transaction, and w_i is the binary intervention decision.

5. EXPLAINABLE AI (XAI) & BILINGUAL LOCALIZATION

To build user trust and satisfy the Responsible AI mandates of Section 14, GenCash rejects opaque decision-making. Every ML recommendation generated by the engine is paired with deterministic, human-readable explanations.

Rather than relying on unconstrained generative hallucination, explanations are generated via bounded semantic templates parameterized by the top feature attributions of the model. Furthermore, because accessibility is paramount in the Bangladesh financial ecosystem, every insight is rendered natively in both **Bengali (বাংলা)** and **English**:

- *English Explanation*: "Recommended because 42% of your past 30-day expenditures were in grocery retail. Claiming this QR deal saves approximately $\text{৳}150$ relative to your standard checkout."
- *Bengali Explanation (বাংলা)*: "আপনার বিগত ৩০ দিনের মোট খরচের ৪২% গ্রোসারি ও কেনাকাটায় হয়েছে বলে এআই এই অফারটি সাজেস্ট করেছে। কিউআর পেমেন্টে এই অফারটি ব্যবহার করলে আপনার প্রায় $\text{৳}১৫০$ সাশ্রয় হবে।"

6. PROTOTYPE IMPLEMENTATION & VERIFICATION

The complete GenCash platform has been engineered into a fully functioning, end-to-end prototype comprising three interdependent subsystems:

6.1 FastAPI Transactional Core

The backend (backend/) provides production-grade endpoints handling atomic P2P Send Money (POST /api/transactions/send), Mobile Recharge (POST /api/transactions/recharge), Merchant QR Payments (POST /api/transactions/payment), Cash Out (POST /api/transactions/cashout), and Bank Add Money (POST /api/transactions/add-money). State integrity is protected through database row-level locking (with_for_update) and double-entry balance validation, preventing race conditions or fraudulent negative balances.

6.2 Customer Mobile Application

Built with React Native and Expo SDK 57 (mobile/), the client interface features:

- **AI Hub & PFM Dashboard**: Real-time aggregation of Money In versus Total Outflow, date-range filtering (Today, This Week, This Month, Last Month, All Time), and exact category expenditure breakdowns.
- **Interactive "Ask AI" Prompting**: Natural language query chips enabling users to request instantaneous expenditure calculations (e.g., "Recharge this month", "Money in this week").
- **Predictive Utility Cards**: Automatic notification of upcoming recurring expenses (e.g., DESCO electricity bills, recurring data recharge cycles) with single-tap payment routing.

6.3 Operator Campaign Simulator Portal

Accessible at /simulator, this web interface allows growth managers to simulate audience segment sizes, configure incentive values, view predicted uplift curves, and observe budget efficiency gains prior to deployment.

7. RESPONSIBLE AI, SECURITY, AND GOVERNANCE

Financial machine learning systems carry direct economic consequences. GenCash enforces the governance principles outlined in Section 14 of the hackathon playbook:

- **Privacy by Design**: Zero customer PII is utilized during model training or inference; all synthetic profiles use hashed pseudonyms.

- **Non-Discriminatory Parity:** Model feature sets omit demographic, gender, and religious indicators, ensuring parity across socio-economic strata.
- **Safety Guardrails:** The AI engine operates strictly in an advisory and ranking capacity. Critical operations (funds transfers, account status changes, PIN resets) require explicit human confirmation via PIN verification and can never be executed autonomously by the AI layer.

8. VALIDATION RESULTS & PATH TO PRODUCTION WITH UPAY

Validation was conducted against a reserved test set of 2,500 synthetic customers over 60 days of transactional activity. The Next-Best-Offer propensity classifier achieved an offline **ROC-AUC of 0.842**, while the uplift allocation strategy demonstrated an estimated **31.4% reduction in subsidy waste** compared to random broadcast targeting.

Table 3: Simulated Performance Comparison Against Baseline Marketing Strategy

Targeting Strategy	Total Campaign Cost	Incremental Transactions	Net Return on Campaign (ROI)
Broad SMS Broadcast	৳ 100,000	1,420	1.42x
Heuristic RFM Rule Filtering	৳ 65,000	1,680	2.58x
GenCash Uplift Optimization	৳ 48,500	2,240	4.61x (+78.6% efficiency)

Post-Hackathon Integration Roadmap

To transition GenCash from hackathon prototype to production reality within upay's enterprise infrastructure:

1. **Stage 1 (Shadow Pipeline):** Deploy the NBO and Anomaly models in passive shadow mode alongside upay's existing notification engine to record baseline prediction telemetry without customer exposure.
2. **Stage 2 (Controlled Pilot):** Execute a small-scale A/B validation test on 5% of opt-in retail customers to verify true incremental transaction lift and app engagement metrics.
3. **Stage 3 (Core Integration):** Connect model inference endpoints to upay's transactional switch using gRPC microservices with real-time feature streaming.

9. CONCLUSION

GenCash illustrates that Artificial Intelligence in Mobile Financial Services achieves its highest value when applied directly to fundamental consumer and operator pain points. By merging client-side spending empowerment with server-side uplift optimization, GenCash establishes a scalable, explainable, and responsible foundation for the next generation of financial technology in Bangladesh.

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